



Integrated system and control design of a one DoF nano-metrology platform



Rudolf Saathof^{1,*}, Markus Thier¹, Reinhard Hainisch, Georg Schitter

Automation and control institute (ACIN), Vienna University of technology, Gusshausstrasse 27-29, Vienna 1040, Austria

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ABSTRACT

Mechanical vibrations and precision of conventional positioning systems are limiting factors for using nano-metrology tools directly in production environments. Vibrations cause relative motion between workpiece and inspection tool, which distorts measurements at the nanometer level. To enable robot based in-line nano-metrology, this paper proposes a metrology platform that is mounted on a robot arm and maintains a constant and precise relative distance to the workpiece by means of a control loop. This paper presents the mechatronic system design of a 1 degree of freedom (DoF) metrology platform for tracking a vibrating sample in the sub-nanometer range. By incorporating control relevant requirements in the mechanical and electrical design, which is supported by a dynamic error budgeting analysis, the implementation of a high bandwidth feedback loop is enabled. The metrology platform consists of a 1 DoF Lorentz actuator with gravity compensator, a low stiffness flexure-based guiding mechanism and a moving mass of 4 kg with high structural resonance frequencies. A high-bandwidth PD based controller that utilizes the signal of an interferometer is implemented for feedback control. Experiments show a tracking error of 4 nm RMS when exposing the sample under test to on-site measured vibrations, which complies with the dynamic error budgeting analysis. This demonstrates viability of the implemented mechatronic design for in-line metrology applications requiring sub-nanometer precision.

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1. Introduction

Nano metrology will be an important tool for future production lines to improve the surface quality of high-end products. Nano metrology tools such as atomic force microscopes (AFMs) and white light interferometers (WLI) can measure structural properties with nm accuracy [8]. By directly monitoring product quality and tool wear at nm level, defect rate and product quality could be improved. This paper proposes robot based in-line metrology as a versatile method to realize this.

Due to limited precision of robotic platforms and sensitivity of nano-metrology tools to environmental vibrations, in-line nano-metrology is not implemented in conventional manufacturing. Industrial positioning robots have position accuracy of only 10 μ m or worse, which is insufficient for nano-metrology. Besides, industrial environments usually are populated by machines and operating personnel [19]. These cause workpiece vibrations, which distort the measurements at the nanometer level, by causing a deviating distance between metrology tool and workpiece. Limited pre-

cision and vibrations urge for the application of precision positioning and vibration isolation techniques. At present, nano-metrology tools are supplied with passive or active vibration isolation tables [21]. Passive vibration isolation tables have high mass and low suspension stiffness, such that vibrations above the suspension frequency are attenuated [1,14]. To reduce mass and volume, active vibration isolation is applied [7,22]. Active vibration isolation tables are equipped with sensors such as accelerometers, which are used in feedback operation. Nano-metrology using these vibration isolation systems require to remove a workpiece from the production line and later insert it again, which is cost and time inefficient. Due to this lack of versatility, nano-metrology tools are hardly used within production lines.

Instead of using vibration isolation tables, correction of vibration influences has been proposed by post-processing of the AFM image [16]. The disturbance between workpiece and AFM is measured by a high resolution distance sensor. This measured disturbance is used to compensate the AFM-image afterward. The disadvantage of this method is the small linear range of the AFM compared to the large required stroke to compensate the disturbance. Alternatively, for an optical measurements device, vibrations are reduced by moving an optic in feedback operation [11]. Hence,

* Corresponding author.

E-mail addresses: saathof@acin.tuwien.ac.at, thier@acin.tuwien.ac.at (R. Saathof).

¹ Both authors equally contributed to this paper.

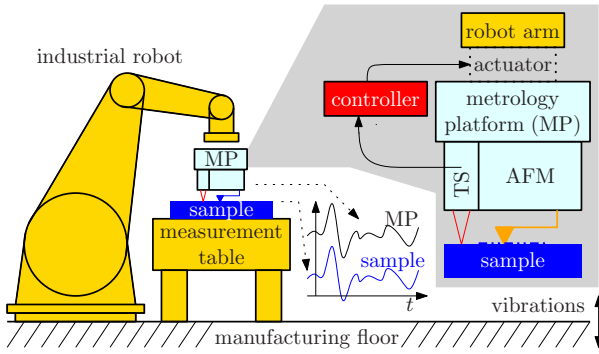


Fig. 1. Metrology platform (MP) concept of robot based nano-metrology. The distance Δz between measurement platform and workpiece is measured by a tracking sensor (TS) and is kept constant using a control loop. Hence, an AFM can measure the nano-topography of the workpiece.

these methods need to be improved and be made applicable for tactile (AFM) measurements.

This paper proposes a dedicated metrology platform design (Fig. 1) for in-line metrology applications which isolates the inspection tool measurement from environmental vibrations by tracking the sample under test. The focus of this paper is on presenting a holistic mechatronic design approach of the metrology platform that considers control relevant requirements already in the mechanical and electronic design step to achieve a high tracking performance. By utilizing a low-stiffness, flexure guided mechanical design with high structural resonance frequencies, zero-stiffness actuation with gravity compensator and a high bandwidth feedback loop, tracking in the nanometer range is demonstrated.

Section 2 presents the system architecture and requirements. The platform design is presented in Section 3 and the system dynamics is modeled in Section 4. Section 5 shows the design of a Proportional Differential (PD) based platform controller that accounts for the high frequency mode shapes of the mechanical plant. Dynamic error budgeting is applied to estimate the overall system performance and the performance on component level, as well as to identify possible bottlenecks of the mechatronic design. Experimental results are presented in Section 6 demonstrating viability of the used approach and tracking of a vibrating sample in the sub-nanometer range.

2. System architecture and requirements

The proposed metrology platform is equipped with a nano-metrology tool, vibration isolation and a high precision positioning system. As such, it can be brought in close vicinity of the workpiece by the robot arm, after which precise positioning is applied. Instead of isolating both workpiece and metrology tool from ground vibrations, the metrology platform keeps a constant distance between workpiece and metrology tool [5,15,20], using a feedback loop.

The high precision positioning system (see Fig. 1) consists of a Lorentz actuator and a high precision tracking sensor measuring the distance between workpiece and metrology platform. After the metrology platform is moved by the robot arm to its intended location, actuator and sensor are used for precision positioning and active vibration tracking using the feedback loop.

2.1. Platform requirements

The intended inspection tools feature single nanometer resolution. Although larger tracking errors compromise the resolution of the measurements, features on the workpiece drive the tracking

Table 1
Measurement platform requirements.

Requirement	Value	Unit
Mass	< 5	kg
Minimum stroke	100	μm
First resonance	> 2	kHz
Residual vibration level (p-v)	< 50	nm
Vibration level to be corrected (p-v)	6	μm

error requirement. A tracking error smaller than 50 nm peak-to-valley in average vibration conditions is specified to identify the features of interest. High bandwidth tracking control is therefore utilized to keep the required residual tracking error.

To realize such a mechatronic system, dedicated design-for-control need to be applied to realize first resonance frequencies well above the desired control bandwidth [17]. Therefore, the platform must be stiff and light weight for handling convenience and limited power consumption of the actuator (Table 1). In addition, minimum weight and maximum stroke are required to comply with the load capacity and limited precision of industrial robots.

2.2. Vibration spectrum

Performance of a vibration isolation system depends on the combination of the mechatronic system and the vibration spectrum. The down-flowed requirements for the mechatronic system design, such as control bandwidth, are derived by assessing a vibration signal, which is measured at a production site. These measurements are presented in Fig. 2 are done at Flubetech S.L. (08227 Terrassa (Barcelona), Spain), which is a potential user as micro and nano coating manufacturer. Vibrations were measured at a workpiece support table, using a Geophone (Type: SM-11, Input/Output Inc., Netherlands).

The recorded time signal of Fig. 2 is separated in time slots with impulses (gray area) and time slots with continuous signal. The recorded time signal has the following characteristics:

- The continuously present vibrations are caused by e.g. production machines. These parts are further separated in time slots of 1 s. The peak-to-valley value in the time slots of 1 s is 1.3 μm . As the statistical properties of these time traces are similar, this part of the signal is assumed to be stationary. The power spectral densities (PSDs) of these time slots are averaged and shown in Fig. 3. Significant vibrations at 25, 50 and 100 Hz are observed.
- Impulses are caused by e.g. operating personnel. As this time trace is too short to determine statistical parameters, they cannot be treated as a stationary process. Rather, these impulses are assumed to be representative with respect to their amplitude and spectral content. The peak-to-valley value is at the micrometer range.

These characteristics provide the following 3 criteria. In case of characteristic (i) the continuous vibrations should be reduced to a level that proper measurements can be done requiring a residual vibration level below 50 nm peak to valley (p-v). This residual vibration level would allow measurements with a standard AFM. The bandwidth requirement is derived using the PSD of Fig. 3 and the mechatronic system design of Sections 3 and 5 in Section 5.4. For characteristic (ii) the vibrations should be reduced to a level that they cannot damage the metrology tool or the workpiece requiring a residual vibration level below 1 μm .

3. System design

For validation and testing purposes, the metrology platform is installed in a rigid mechanical frame that replaces the robot arm.

Table 2
Actuator properties.

Specification	Value	Unit
Windings	164	~
Inductance	0.76	mH
Resistance	2.6	Ω
Effective length	17.4	m
Wire diameter	0.4	mm
Magnetic field B_r	1.2	T
Gravity compensation force	40.5	N

The measurement system is placed on a separate mechanical frame to resemble the real scenario. The schematic of the implemented laboratory setup is depicted in Fig. 4.

3.1. Metrology platform

The metrology platform is represented by an aluminum box with 10 mm thick walls with a mass of 4.0 kg. This mass is chosen because it does not exceed the typical load capacity of 5 kg of industrial robots, but it does provide a representative load of the metrology instruments, tracking sensors and carrying mechanical structure. As the metrology platform has a closed box construction, it is lightweight and stiff [18].

The metrology platform has two elastic hinges to enable straight movement in vertical (z) direction and constraint the other 5 DoFs [4]. To provide stiffness to these 5 DoFs and to prevent unwanted structural modes caused by the flexure mechanism, a notch type linear spring is implemented. It consists of a leaf spring which is clamped between two solid bodies to provide two elastic joints. The stiffness in the z-direction is designed to be as low as possible.

To determine a reasonable requirement on the resonance frequency of the suspension mode, vibration measurements were performed on an ABB IRB140 M2000 robot. It showed significant vibrations of the end-effector above 10 Hz when exciting the work floor by impulses. Hence, the suspension resonance frequency should be smaller than 10 Hz to prevent vibrations to propagate from the robot arm to the metrology platform. Fig. 5 shows the realization of the metrology platform embedded in an experimental setup.

3.2. Actuator design

The used Lorentz actuator with integrated gravity compensator is shown in Fig. 6 [2]. The actuator is similar to a voice coil actuator, having a circular coil positioned within a ring of magnets. The coil has 164 windings and the mover ring magnets a magnetic field of 1.2 T. This creates a Lorentz force that is controlled by the current. To keep the metrology platform in its working position, the actuator has an integrated gravity compensator, which is tuned on the weight of the metrology platform. As such it avoids a continuous current that produces heat. As vibrations could also enter the metrology platform from the robot arm, the actuator provides vibration isolation. Vibration isolation from the robot arm is provided by (i) no mechanical connections between stator and mover and (ii) positioning the mover in the working point of the stator with low stiffness [2].

The actuator is 3D printed. It has neodymium magnets (NdFeB 37). Further actuator specifications are mentioned in Table 2.

3.3. Electronic hardware

The current in the actuator is provided by a power amplifier (type MP38CL, Apex Microtechnology, USA). The amplifier is current-controlled using a PI-controller implemented on the FPGA

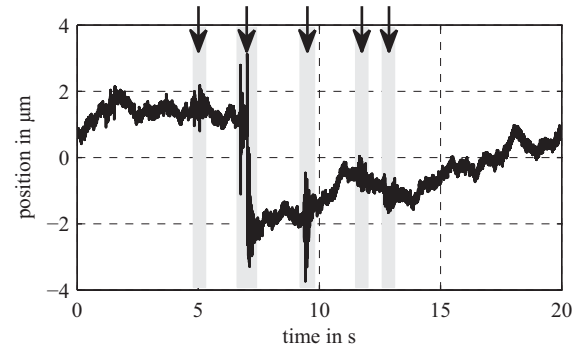


Fig. 2. Vibration signal measured in a production environment. It is recorded with a Geophone on a workpiece support table. The continuous vibration signal has a peak-to-valley value of 400 nm, the impulses have an amplitude of 6 µm maximum (gray area with arrows).

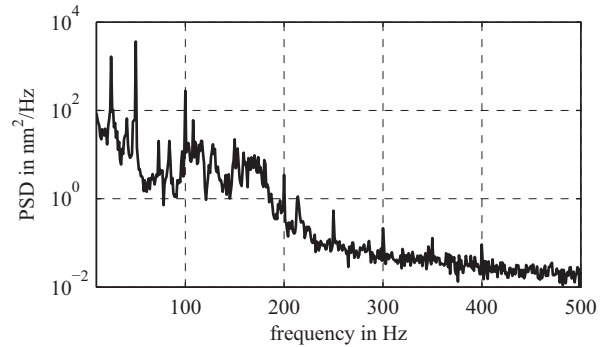


Fig. 3. Averaged power spectral density of the continuous vibration signal of Fig. 2 (the non-gray area).

of the rapid prototyping system (DS1005, dSpace GmbH, Germany) at a bandwidth of 10 kHz with a noise level of 3.5 mA-rms. Also the tracking controller is implemented on this system, which runs at a sample frequency of 20 kHz.

3.4. Vibration source

In the laboratory setup, the loudspeaker emulates the on-site measured vibrations (see Fig. 2). The current for the loudspeaker is provided by a feed forward signal using an inverse model of the loudspeaker.

3.5. Sensor system

The tracking sensor consists of a heterodyne Michelson interferometer (Agilent 10898A, Agilent Technologies, USA), which has a resolution of 1.25 nm. As the Michelson interferometer provides a differential measurement, one laser beam goes towards the measurement platform, and the other arm to the loudspeaker. To avoid structural loops from the actuator stator to the workpiece, the interferometer is placed on a separate table. This provides an environment equivalent to the targeted production environment, since the robot arm would hold the metrology platform above a work piece.

4. Model

For controller design and comparison, a electromechanical model is devised.

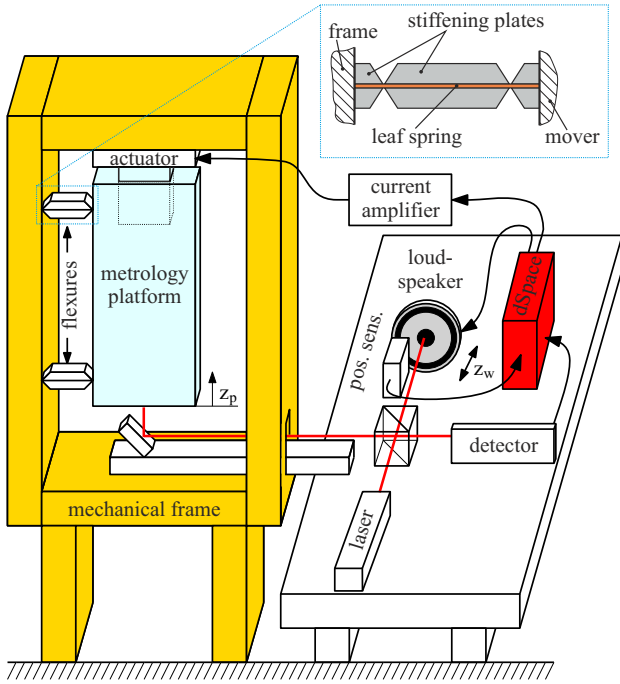


Fig. 4. Schematic of the laboratory prototype. A heterodyne Michelson interferometer measures the relative distance between metrology platform and workpiece. Vibrations z_w are emulated by a loud speaker and are measured by a position sensor (pos.sens.) for validation. A current amplifier drives a Lorentz-actuator with gravity compensation in vertical direction z_p .

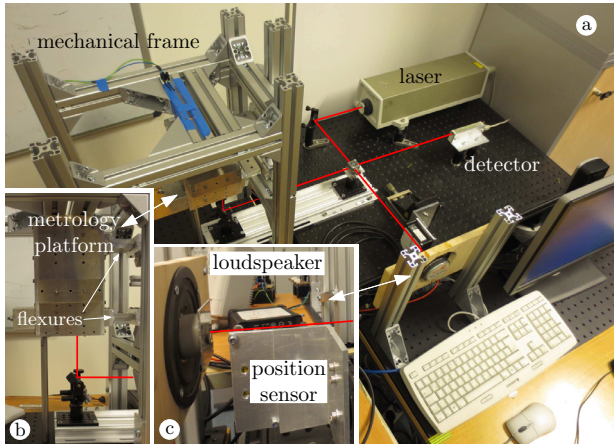


Fig. 5. Laboratory prototype to evaluate vibration mitigation. (a) Overview of the setup with the laser interferometer, (b) metrology platform with the flexures, (c) loudspeaker to emulate a vibration source.

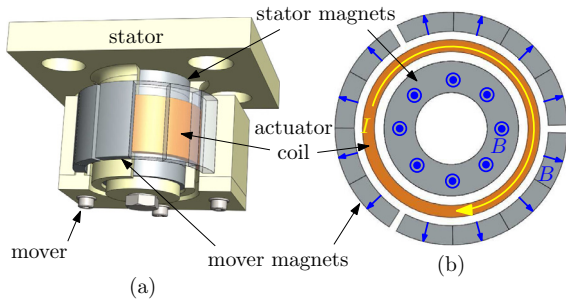


Fig. 6. Schematic of the Lorentz actuator with integrated gravity compensator [2]. (a) 3D view with cross sections, (b) magnets and coils.

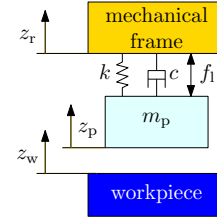


Fig. 7. Simplified system model of the metrology platform, considering the metrology platform as rigid body with mass m_p . The suspension mechanism is modeled by the spring k and damper c . The actuation is provided by a Lorentz force f_l .

4.1. Nominal mechanical model

The model depicted in Fig. 7 has mass m_p , stiffness k a damping coefficient c and a Lorentz force f_l originating from the actuator. Stiffness $k = k_h + k_{gc}$ consists of the stiffness of the elastic hinges k_h and the actuator k_{gc} , which is derived in Section 4.3. The blocks represent bodies with positions z_p , z_r and z_w describing the position of the measurement platform, the robot arm and the workpiece respectively.

$$m_p \ddot{z}_p(t) + k(z_p(t) - z_r(t)) + c(\dot{z}_p(t) - \dot{z}_r(t)) = f_l(t). \quad (1)$$

By using the Laplace transformation and rewriting it yields:

$$Z_p(s) = [F_l(s) + Z_r(s)P_r(s)]P_n(s), \quad (2)$$

with

$$P_n(s) = \frac{1}{m_p s^2 + cs + k}, \quad (3)$$

$$P_r(s) = k + cs. \quad (4)$$

in which P_n represents the nominal metrology platform dynamics and P_r the influence of robot arm vibrations to the force acting on the metrology platform.

4.2. Higher order model

As the entire metrology platform is a relative complex structure, the mechanical model could be expanded with higher resonance modes. The modal description of this expanded model is indicated by P_m :

$$P_m(s) = P_n(s) + \Delta P_m^n(s), \quad (5)$$

with $\Delta P_m^n(s)$ denoting flexible body modes up to mode n of interest. To determine $\Delta P_m^n(s)$, the entire system is modeled using mass matrix $\mathbf{M}$, stiffness matrix $\mathbf{K}$, damping matrix $\mathbf{D}$, force vector $\mathbf{F}$ containing external forces, and $\mathbf{z}$ the platform position vector:

$$\mathbf{M}\ddot{\mathbf{z}}(t) + \mathbf{D}\dot{\mathbf{z}}(t) + \mathbf{K}\mathbf{z}(t) = \mathbf{F}(t). \quad (6)$$

By modal analysis, this equation can be written as a superposition of individual modes r [13]:

$$\frac{Z_i(s)}{F_j(s)} = \sum_{r=1}^n \frac{\phi_{ir}\phi_{jr}}{m_r(s^2 + 2\zeta_r\omega_r s + \omega_r^2)} \quad (7)$$

where the displacement z_i is the displacement of node i and the external force acts on node element j . The modal mass m_r and the modal damping ζ_r are obtained by:

$$m_r = \phi_r^T \mathbf{M} \phi_r, \quad \zeta_r = \frac{\phi_r^T \mathbf{D} \phi_r}{2\omega_r}, \quad (8)$$

in which the eigenvectors ϕ_r are normalized to yield an identity modal mass matrix and ω_r are the resonance frequencies such that:

$$\mathbf{K}\phi = \mathbf{M}\phi\omega^2, \quad (9)$$

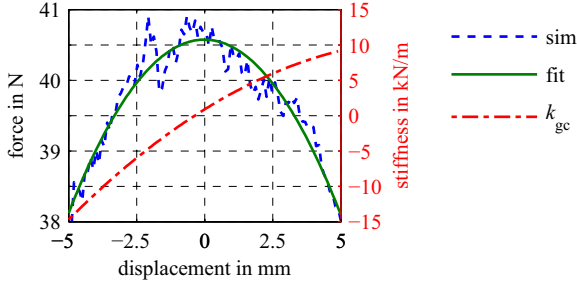


Fig. 8. Simulation of the gravitation force and stiffness using FEMM [10] (sim). The label fit is a second order fit of the FEMM simulation result. Actuator stiffness k_{gc} is obtained by a second order fit of the derivative of (sim).

in which

$$\omega^2 = I \begin{bmatrix} \cdots & \omega_r^2 & \cdots \end{bmatrix}^T, \quad \phi = \begin{bmatrix} \cdots & \phi_r & \cdots \end{bmatrix}^T, \quad (10)$$

in which I is the identity matrix. By using the suspension mode described in Eq. (3), Eq. (5) and by defining $a_r := \phi_m, r\phi_j, r/m_r$, transfer function P_m can be written as:

$$P_m(s) = P_n(s) + \sum_{r=2}^n \frac{a_r}{(s^2 + 2\zeta_r\omega_r s + \omega_r^2)}, \quad (11)$$

Assuming masses m_r and resonance frequencies ω_r can be precisely determined, parameters k , c , a_r and ζ_r can be identified using a system identification. This system identification is based on manual fitting of these parameters using a measured frequency response as indicated in Section 4.6.

4.3. Actuator model

The relation between the Lorentz force F_l and the current i_a generated by the actuator is

$$F_l = \underbrace{K_m}_{Bln} I, \quad (12)$$

in which B is the magnetic field, l the length of the coil that is effectively in the magnetic field, n the number of windings.

In addition, the gravitation compensation force has been obtained using the Finite Element Method Magnetics (FEMM) package, with an axisymmetric model of the actuator [10]. The gravitational force in Fig. 8 shows some noise, due to the changing mesh of the model. Hence, it is fitted with a second order polynomial using least squares. Fig. 8 shows an optimum of 40.5 N compensation force. The magnetic stiffness of the gravitation compensator k_{gc} results from the position-force dependency of the relative position of the magnets. The stiffness is determined by the position derivative of the simulated force and fitted with a second order polynomial.

The suspension stiffness k is mainly determining the transmission of vibrations from the stator to the mover of the actuator as indicated by Eq. (4). Because minimum vibration transfer is desired, the suspension stiffness k should be close to zero, such that $P_r \approx 0$. In Fig. 8, $k_{gc} = 0$ at a displacement of 0 mm. Hence, this is the ideal working point of the actuator. Within a range of ± 0.5 mm around the working point the stiffness varies with ± 1.5 kN/m.

4.4. Electrical model

The current is controlled using an FPGA through the circuit that can be described by:

$$V_o = G_c(s)V_i - V_m = \frac{R_s}{L_c s + R_c + R_s} V_i - V_m. \quad (13)$$

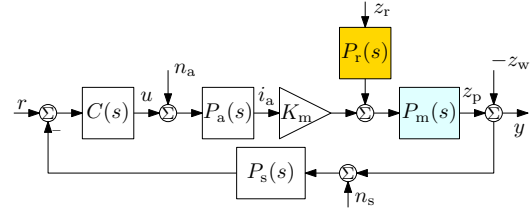


Fig. 9. Block diagram of the control loop. y is the control variable, and u the control signal. Blocks $P_m(s)$ and $P_r(s)$ represent the mechanical plant and the connection to the robot respectively. $P_a(s)$ and K_m are current amplifier and the motor constant. $P_s(s)$ and C denote the sensor dynamics and the controller. Vibrations of the robot z_r and z_w are modeled as a system and an output disturbance respectively. Other noise sources n_a and n_s originate from the amplifier and sensor.

in which G_c is the transfer function from input voltage V_i to the output voltage V_o , R_c and L_c are the resistance and the inductance of the actuator coil, and V_m denotes the back electromotive force (EMF), which is considered as disturbance. The output voltage V_o is proportional to the current by the shunt resistor R_s . By controlling the output voltage via controller $C_c(s)$, current control of the actuator is realized:

$$C_c(s) = k_p + k_i/s, \quad (14)$$

with k_p and k_i the proportional and integral gain respectively. By choosing $k_p = L_c 2\pi f_c / R_s$ and $k_i = (R_c / R_s + 1) 2\pi f_c$, pole-zero cancellation is applied to the pole originating from $L_c s + R_c + R_s$. Hence, the closed loop transfer function has a first order low pass characteristic:

$$P_a(s) = \frac{C_c(s)G_c(s)}{1 + C_c(s)G_c(s)} = \frac{1}{s/(2\pi f_c) + 1}, \quad (15)$$

with f_c denoting the bandwidth of the current control loop. A bandwidth of $f_c = 10$ kHz has been implemented to enable high bandwidth position control.

4.5. Complete model

Using the block diagram of Fig. 9 the control variable $y = z_p - z_w$, is defined. It represents the distance between workpiece and measurement platform that is measured by the interferometer. The control signal u is provided by the feedback controller and fed to the power amplifier with dynamics $P_a(s)$, resulting in the control current i_a . Hence, the complete plant to be controlled is defined as

$$P(s) = U(s)/Z_p(s) = K_m P_a(s) P_m(s) P_s(s), \quad (16)$$

with P_s the sensor dynamics which is modeled as a unity gain.

4.6. System modeling

The system model is fitted to a measured frequency response by a logarithmic sine sweep from 1 to 20 kHz. The frequency response is recorded by the dSpace platform using the lock-in principle [9]. The bode plot of Fig. 10 shows the measured frequency responses and the fitted model.

This fit is used to estimate the characteristics that are expressed in Eq. (12), Eq. (3) and Eq. (11). The nominal mass-damper-spring model dominates up to 2 kHz with a mass of m 4.0 kg, a stiffness of 2.0 kN/m and a damping d of 1 Ns/m, and the characteristics of the higher frequency resonances are presented in Table 3. The motor constant K_m of the actuator is 1.24 N/A.

5. Controller design

The control strategy is to keep a constant relative distance y between workpiece and measurement platform, forming a distur-

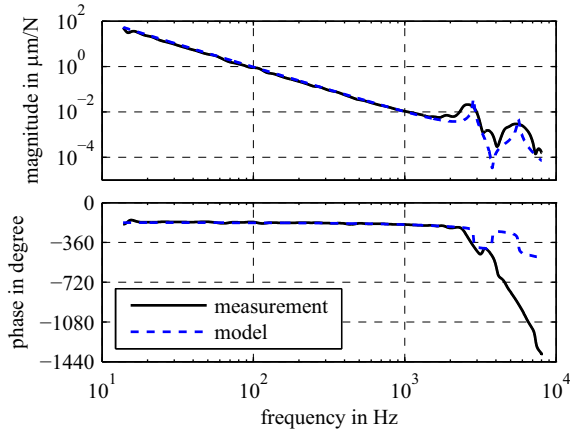


Fig. 10. Measured and fitted transfer function. The mass-spring-damper character goes up to 2 kHz. Characteristics of the higher resonances are presented in Table 3. In addition, a sampling delay of 50 μ s is modeled.

Table 3
Coefficients of the fitted high order dynamics of $P_m(s)$ (Eq. 11).

Mode	m in kg	k in kN/m	c in Ns/m
P_n	4.0	2.0	1
P_m^2	f_r in Hz	a_r in m/N	ζ_r
P_m^2	$2.9 \cdot 10^3$	-0.15	$5.3 \cdot 10^{-3}$
P_m^3	$5.6 \cdot 10^3$	-0.10	$7.4 \cdot 10^{-3}$

bance correction problem. Taking into consideration a high targeted control bandwidth and the plant exhibiting double integrator behavior over a large frequency range, a PD based controller is selected as tracking controller. Please note that using additional integral control can further improve the disturbance rejection in the lower frequency range. However, at the same time it results in a larger control action and more heat dissipation in the power amplifier and the actuator. Thus additional integral control is not implemented to avoid overheating due to high control bandwidths. In the following section, the controller parametrization is presented and the controller bandwidth is derived.

5.1. Tamed PD controller

The PD controller is parametrized as:

$$C_{pd}(s) = K_p + \frac{K_d s}{1 + K_t s}. \quad (17)$$

The proportional gain K_p is used to provide the required controller gain. Due to the double integrator of the plant, the phase is -180° , which would yield to instability at the cross over frequency. Therefore a derivative gain K_d is used to lift the phase around the crossover frequency. To overcome high-frequency noise amplification, the derivative control action is tamed by using a first order low pass characteristic indicated by K_t . The controller parameters are set using [12]:

$$K_p = m_p \omega_c^2 / \sqrt{\omega_d}, \quad (18)$$

$$K_d = m_p \omega_c, \quad (19)$$

$$K_t = 1 / (\sqrt{\omega_d} \omega_c), \quad (20)$$

with ω_c as the crossover frequency and ω_d denoting a tuning parameter to specify the bandwidth of the differentiator. To achieve

Table 4
Parameters of the implemented Notch filters.

Parameter	$\omega_n/2\pi$ (Hz)	ζ_n	d
$C_{n,1}(s)$	$2.8 \cdot 10^3$	0.75	120
$C_{n,2}(s)$	$3.5 \cdot 10^3$	0.25	100

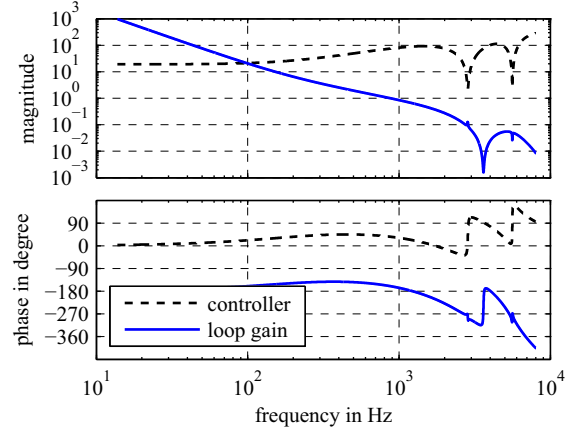


Fig. 11. Model of the PD controller with notch filters and the loop gain $L(s) = G(s)C(s)$.

a good trade-off between robustness and performance for spring-mass systems, a ω_d of 9 is typically chosen. As additional filters are needed to account for the flexible modes at higher frequencies, $\omega_d = 24$ is selected for implementation to increase the robustness of the PD controller around the cross-over frequency.

5.2. Notch filters

In order to specifically suppress the second and third resonance frequency of 2.9 kHz and 5.6 kHz Notch filters are used. Notch filters are beneficial with respect to phase margin compared to low pass filters that would shift the phase around the resonances to obtain a stable closed loop system. The following parametrization is used:

$$C_n(s) = \frac{s^2 + 2(\zeta_n/d)\omega_n s + \omega_n^2}{s^2 + 2\zeta_n\omega_n s + \omega_n^2}, \quad (21)$$

in which the notch frequency ω_n is set to the resonance frequencies ω_r of the metrology platform (Eq. 7), ζ_n is the logarithmic width of the notch and d the multiplicative depth. These values are tuned to suppress the resonances, as shown in Table 4 and Fig. 11. As can be seen in Fig. 11, the phase lead from the derivative action is maximum at around 400 Hz, instead of around the crossover frequency. This is due to the notch filters that also reduce the phase at the crossover frequency. The PD controller $C_{pd}(s)$ and the notch filters $C_{n,1}(s)$, $C_{n,2}(s)$ have been added together:

$$C(s) = C_{pd}(s)C_{n,1}(s)C_{n,2}(s), \quad (22)$$

to form the feedback controller $C(s)$.

5.3. Dynamic error budget

To compute the residual vibration error of the mechatronic system presented so far dynamic error budgeting is used [6]. The power spectral density of the system output $PSD_y(\omega)$ and the con-

Table 5
PSD's of the tracking sensor and the power amplifier.

PSD	Value	Unit
PSD_{ts}	450	pm^2/Hz
PSD_a	2500	$\mu\text{N}^2/\text{Hz}$

troller output $PSD_u(\omega)$ is determined by:

$$PSD_y(\omega) = |S(\omega)|^2 PSD_v(\omega) + |S_{y,r}(\omega)|^2 PSD_r(\omega) + |S_{y,a}(\omega)|^2 PSD_a(\omega) + |S_{y,s}(\omega)|^2 PSD_{ts}(\omega), \quad (23a)$$

$$PSD_u(\omega) = |S(\omega)|^2 PSD_a(\omega) + |S_{u,r}(\omega)|^2 PSD_r(\omega) + |S_{u,y}(\omega)|^2 (PSD_v(\omega) + PSD_{ts}(\omega)), \quad (23b)$$

with PSD_v and PSD_r the vibration spectrum of the workpiece support table (Fig. 3) and the robot arm respectively, and PSD_a and PSD_{ts} the noise spectra of the amplifier and the tracking sensor (Table 5). S is the sensitivity function and $S_{i,j}$ are the sensitivity functions from input j and output i defined as:

$$S(\omega) = \frac{1}{1 + P_m(\omega)K_m P_a(\omega)C(\omega)P_s(\omega)} \quad (24a)$$

$$S_{y,r}(\omega) = S(\omega)P_m(\omega)P_r(\omega) \quad (24b)$$

$$S_{y,a}(\omega) = S(\omega)P_m(\omega)K_m P_a(\omega) \quad (24c)$$

$$S_{y,s}(\omega) = T(\omega) = S(\omega)P_m(\omega)K_m P_a(\omega)C(\omega)P_s(\omega) \quad (24d)$$

$$S_{u,y}(\omega) = S(\omega)C(\omega)P_s(\omega) \quad (24e)$$

$$S_{u,r}(\omega) = S(\omega)C(\omega)P_s(\omega)P_m(\omega)P_r(\omega). \quad (24f)$$

From these sensitivity functions, the influence of transmissibilities $S_{y,r}$ and $S_{u,r}$ are neglected for the following reasons: First, as explained in Section 4.3, $P_r \approx 0$ due to zero stiffness suspension. Hence, the transmission of PSD_r is negligibly small compared to the transmissibility of PSD_v . Second, for this reason, no stimulus for emulating robot vibrations is included in the laboratory setup. Third, the vibration level of PSD_r in the laboratory is so low that its contribution to the performance is negligibly small and could not be observed. Hence, $S_{y,r}$ and $S_{u,r}$ are not taken into account for our setup, but are provided here for completeness. The root mean square (RMS) value of the remaining vibration spectrum up to a frequency f_e is computed with:

$$y_{\text{rms}} = \sqrt{\int_0^{f_e} PSD_y(\omega) d\omega} \quad (25a)$$

$$u_{\text{rms}} = \sqrt{\int_0^{f_e} PSD_u(\omega) d\omega}. \quad (25b)$$

5.4. Derivation of the bandwidth requirement

According to the requirement, the tracking error should be kept within 50 nm. Taking a confidence interval of 6σ , the RMS error should be below 8.3 nm. Using the dynamic error budget of Eq. (16), and the PSD presented in Fig. 3 as input, the residual tracking error is obtained by Eq. (25a). By sweeping ω_c over a

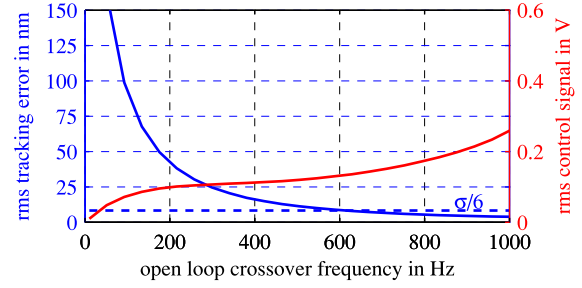


Fig. 12. Simulated performance plot with the error and the control effort as function of the controller bandwidth.

Table 6
Summary of the experimental results.

Parameter	Performance
Peak to valley (continuous)	25 nm
rms position error (continuous)	4.4 nm
Peak-to-valley (peaks)	600 nm
Control bandwidth	850 Hz
Phase margin	17°
Gain margin	2 dB

bandwidth from 10 Hz to 1 kHz the bandwidth dependent residual error and actuator performance is obtained (Fig. 12). At 600 Hz a residual vibration error of 8.3 nm-rms is achieved. In order to boost the performance the bandwidth is set to 850 Hz with a gain and a phase margin of 2 dB and 17° respectively.

5.5. Revisiting noise sources

The cumulative PSD as plotted in Fig. 13 shows that the vibration noise is still the main contributing noise source. Only at frequencies below 10 Hz, the sensor noise is dominant, but this can only be prevented by decreasing bandwidth, or improving the sensor. In addition Fig. 13 shows that the lower frequencies (< 20 Hz) are well suppressed by the PD controller and that an integrator of a PID controller will not significantly reduce the tracking error further. Fig. 14 shows that up to frequencies of 1 kHz controller effort is mainly driven by the vibration level. Above this frequency, sensor noise is the dominant contributing source. To further decrease the RMS value of the control signal, a sensor with less noise would be required.

5.6. Implementation

The controller for 850 Hz bandwidth is implemented on the dSpace system using pole-zero-matching [3], with a sampling frequency of 20 kHz. The implemented control gains are: $K_p = 23.3 \cdot 10^6$, $K_d = 21.4 \cdot 10^3$ and $K_t = 38.2 \cdot 10^{-6}$. The difference of the open loop crossover frequency and complementary sensitivity crossover frequency are clearly visible in Fig. 15. In addition, it shows the suppression of the high frequency resonances by the notch filters.

6. Experiments

To verify the system design, achieved bandwidth and vibration level, the frequency and time responses and are summarized in Table 6. The open loop and closed loop transfer functions are presented in Fig. 15

The sensitivity function in Fig 16 shows the vibration suppression in the frequency domain. The sensitivity crossover frequency is at 490 Hz. Due to the high bandwidth, the gain margin and phase margin are only low, causing a relatively high peak in the

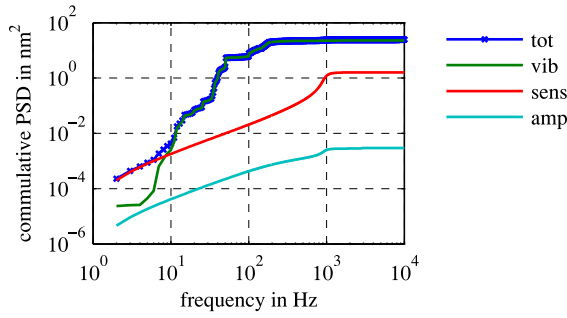


Fig. 13. Computed cumulative PSD of the tracking error from the noise sources (vib = vibrations, amp = amplifier noise, sens = tracking sensor noise and tot = total noise).

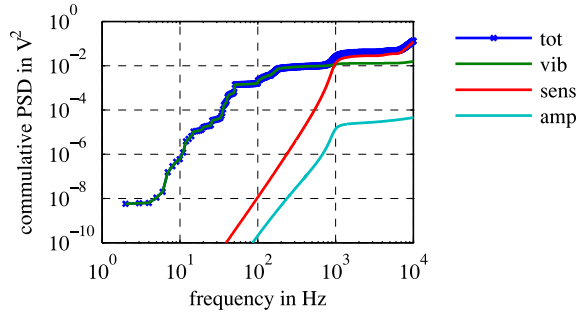


Fig. 14. Computed cumulative PSD of control output from the noise sources (vib = vibrations, amp = amplifier noise, sens = tracking sensor noise and tot = total noise).

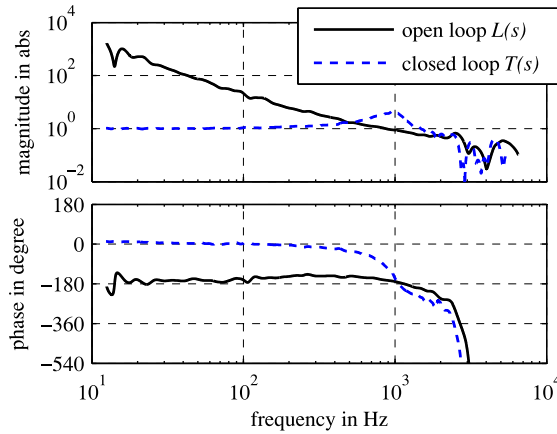


Fig. 15. Measurement of the open and closed loop transfer function. The open loop cross over frequency is 850 Hz.

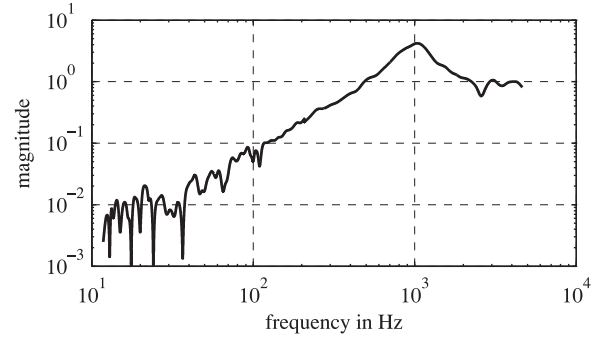


Fig. 16. The measured sensitivity function of the closed loop controlled systems has a bandwidth of 490 Hz.

sensitivity function around 1 kHz. As the vibrational spectrum shows no disturbance at high frequencies, this has no negative effect on the vibration rejection capability.

The time signals are recorded with the controller in Fig. 17. The peak disturbances are reduced from 6 μm peak-to-valley to 600 nm peak-to valley, a 10 fold reduction. The maximum peak-to-valley error found the selected time frames of 1 s is reduced from 1.3 μm to 25 nm as shown in Fig. 17.

In order to verify the dynamic error budget of Section 5.3, PSDs of the results are presented in Fig. 18. The PSD of the original signal (equivalent to Fig. 3) is propagated through the system using Eq. (23b) (a). The results show excellent resemblance of the estimated (blue) and measured (black) residual vibration spectrum. In addition, it shows excellent steady-state suppression. In summary, these results show that the proposed metrology platform can reduce vibrations between the workpiece and metrology tool to a sufficiently low level to enable robot-based in-line nano-metrology.

7. Conclusions

A metrology platform is proposed to enable nano-metrology in a production environment. Instead of isolating vibrations of a workpiece, the platform follows the vibrations of the workpiece using a control loop. A Lorentz actuator with gravity compensation provides the actuation force, passive gravity compensation, and vibration isolation from the robot arm. The metrology platform has high structural stiffness to enable a high control bandwidth.

By implementing a PD controller with two notch filters to suppress the higher resonance frequencies, a bandwidth of 850 Hz is achieved with a residual tracking error of 4.4 nm rms. This shows the feasibility of the proposed solution, which will enable in-line

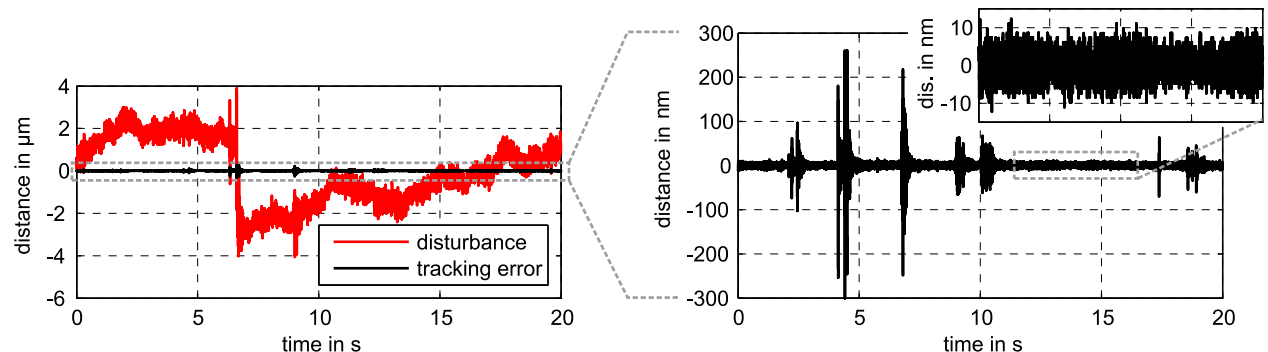


Fig. 17. Measured time signal of the emulated compared to the residual tracking error.

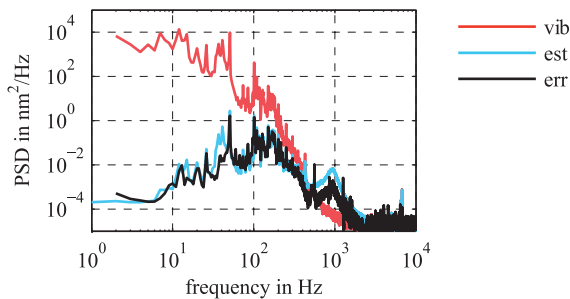


Fig. 18. PSDs compared: the PSD of the emulated vibration source (vib), the estimated PSD of the tracking error (est), and measured PSD of the tracking error (err).

nano-metrology. Ongoing work is extending this concept to 6 DoF, for rejecting vibrations in all spatial directions.

Acknowledgment

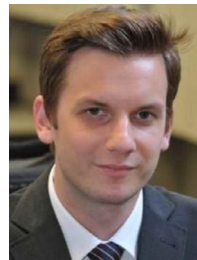
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Rudolf Saathof was post-doctoral researcher at the Automation and Control Institute (ACIN) at TU Vienna (2013–2016). His main research interests are optomechatronic systems for high precision instruments. These include vibration isolation and fine positioning systems, and adaptive optics for laser communication and lithography. He received his master degree in mechanical engineering at Twente University, the Netherlands (2009) and his PhD degree at Delft University of Technology, the Netherlands (2013). Currently he is scientific researcher at the Dutch organization for Applied Scientific Research (TNO) in the Netherlands, Delft at the department for optomechanics.



Markus Thier received his master degree in electrical engineering from Vienna University of Technology in 2012. He is currently working towards his PhD at the Automation and Control Institute (ACIN) at Vienna University of Technology. His primary research interests are on mechatronic systems design, precision engineering, inline metrology, and vibration isolation systems.



Reinhard Hainisch was researcher at the Automation and Control Institute (ACIN) at Vienna University of Technology since 2013. He received a MSc. in Electrical and Biomedical Engineering from Graz University of Technology, Austria (2003) and his PhD degree from Vienna University of Technology, Austria (2015). His primary research interests are in the field of system design and medical engineering. Currently he is metrology developer at Infinion.



Georg Schitter is Professor at the Automation and Control Institute (ACIN) of TU Wien. He received an MSc in Electrical Engineering from TU Graz, Austria (2000) and an MSc and PhD degree from ETH Zurich, Switzerland (2004). His primary research interests are on high-performance mechatronic systems, particularly for applications in the high-tech industry, scientific instrumentation, and mechatronic imaging systems, such as AFM, scanning laser systems, telescope systems, adaptive optics, and lithography systems for semiconductor industry. He received the IFAC Mechatronics best paper award (2008–2011) and 2013 IFAC Mechatronics Young Researcher Award, and served as an Associate Editor for IFAC Mechatronics, Control Engineering Practice, and for the IEEE-Transactions on Mechatronics.